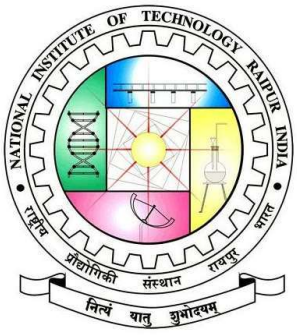
National Institute of Technology Raipur

Department of Information Technology



Minor Project Report

on

AgriVision AI: AI-Powered Crop Health Monitoring and Failure Diagnosis

Detection of crop diseases, pest damage and nutrient deficiencies from a single leaf photograph using computer vision

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Abstract

Crop diseases, pest attacks and nutrient deficiencies are usually recognised only after visible damage has become severe, and farmers often lack timely access to agricultural experts who can distinguish between them. This project presents **AgriVision AI**, an end-to-end decision-support system that diagnoses crop health problems from a single photograph of a leaf. The system validates and pre-processes the image, isolates the leaf from its background using colour analysis for plain-background photographs and a focus-seeded GrabCut algorithm for field photographs, and classifies it into 39 classes covering 14 crops with a MobileNetV3-Large network adapted through transfer learning. Each result is routed into one of four categories (disease, pest, nutrient deficiency or healthy), the proportion of damaged tissue is converted into a 0–100 crop health score and a severity band, and a knowledge-based recommendation engine produces treatment and prevention advice that accounts for severity, growth stage, weather, soil readings, region and the diagnosis history of the field. Confidence is calibrated with temperature scaling, and the system abstains and requests a new photograph whenever confidence is insufficient.

The model was trained on 10,591 laboratory images from PlantVillage and 2,005 field photographs from PlantDoc. On a held-out laboratory test set of 2,270 images it achieves an accuracy of 97.0% and a macro F1-score of 0.970; with abstention, the false diagnosis rate falls to 1.8%. On an independent test set of 236 field photographs, accuracy rises from 19.9% for a model trained on laboratory images alone to 53.8% after field images are included in training, quantifying the gap between laboratory and real-world conditions. The system is delivered as a Streamlit web application, a FastAPI REST service and a native Android application, and generates downloadable PDF crop-health reports. Training requires only a CPU, and a complete diagnosis takes under two seconds on a standard laptop.

Keywords: plant disease detection, computer vision, transfer learning, MobileNetV3, image segmentation, confidence calibration, selective classification, precision agriculture.

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List of Abbreviations

API	Application Programming Interface
CLAHE	Contrast-Limited Adaptive Histogram Equalisation
CNN	Convolutional Neural Network
CPU	Central Processing Unit
ExG	Excess Green index
FDR	False Diagnosis Rate
HSV	Hue, Saturation, Value colour space
KVK	Krishi Vigyan Kendra (agricultural extension centre)
MLP	Multi-Layer Perceptron
PDF	Portable Document Format
ReLU	Rectified Linear Unit
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface

CHAPTER 1

Introduction

1.1 Background and Motivation

Plant pathogens and pests remain one of the largest threats to food security. A global expert survey estimated yield losses of 21.5% for wheat, 30.0% for rice, 22.5% for maize, 17.2% for potato and 21.4% for soybean attributable to pathogens and pests [1]. A large share of these losses is avoidable: most foliar diseases can be contained if they are recognised early and treated correctly. In practice, however, farmers frequently identify problems only once the visible damage has become severe.

Several factors contribute to this delay. Many diseases look alike in their early stages, nutrient deficiencies produce discolouration that closely resembles disease symptoms, pest damage spreads before it is noticed, and manual field inspection is slow and labour-intensive. Access to agricultural experts is limited, particularly for small farms. The consequences compound along a predictable chain: late detection leads to incorrect diagnosis, incorrect diagnosis to improper treatment, improper treatment to reduced yield, and reduced yield to financial loss. Intervention is most effective at the first link of this chain.

Advances in computer vision and the availability of smartphones with good cameras make it feasible to move the first diagnostic step to the farmer. Deep convolutional neural networks have reached very high accuracy on curated plant disease datasets [2, 3, 4], and lightweight architectures make inference practical on modest hardware [5]. Turning a classifier into a useful tool, however, requires more than a class label: the user needs to know how severe the problem is, how reliable the diagnosis is, and what to do next [6, 7].

1.2 Problem Statement

The objective is to design and implement a scalable computer-vision application that allows a farmer to photograph a crop leaf and receive, without waiting for an expert visit, (i) the probable problem and its cause, (ii) whether it is a disease, a pest attack, a nutrient deficiency or a healthy plant, (iii) an estimate of severity, (iv) actionable, context-aware recommendations and (v) a crop-health report, while ensuring that uncertain diagnoses are flagged rather than presented as fact.

1.3 Objectives

1. To build an image pipeline that validates photographs, corrects poor exposure and separates the leaf from both plain and cluttered backgrounds.

2. To train a lightweight deep-learning classifier for 38 crop conditions across 14 crops, together with a class for images that contain no leaf.
3. To route every diagnosis into one of four categories — disease, pest, nutrient deficiency or healthy — and to screen for nitrogen, potassium and iron/magnesium deficiency patterns.
4. To estimate severity as a 0–100 crop health score and a four-level band from the damaged leaf area.
5. To generate recommendations that depend on crop, severity, growth stage, environmental conditions, region and the diagnosis history of the field.
6. To make the system reliable through calibrated confidence and abstention, and to evaluate it on both laboratory and real field photographs.
7. To deliver the system through a web application, a REST API and an Android application, with PDF reports and persistent history.

1.4 Contributions

The main contributions of this work are the following:

- A complete, working diagnosis system that integrates validation, segmentation, classification, severity estimation, nutrient screening, recommendations, reporting and history in a single pipeline, deployed on three interfaces.
- A quantitative measurement of the laboratory-to-field gap on this task: a model trained only on laboratory images reaches 97.4% on laboratory test images but only 19.9% on field photographs; including field photographs in training raises field accuracy to 53.8% while keeping laboratory accuracy at 97.0%.
- A focus-seeded GrabCut segmentation method that isolates the photographed leaf from surrounding foliage by exploiting the camera's depth of field.
- Separate, data-driven abstention thresholds for laboratory-style and field photographs, so that the system trades coverage for reliability where the task is harder.

1.5 Organisation of the Report

Chapter 2 reviews related work and identifies the research gap. Chapter 3 presents the system architecture and technology stack. Chapter 4 describes the methodology: datasets, preprocessing, segmentation, the classification model, abstention, severity estimation, nutrient screening and the recommendation engine. Chapter 5 covers the implementation of the web application, REST API, Android application and reports. Chapter 6 reports experimental results, Chapter 7 discusses them, Chapter 8 lists limitations and future work, and Chapter 9 concludes the report.

CHAPTER 2

Literature Review

2.1 Image-Based Plant Disease Recognition

Early approaches to automated disease recognition relied on hand-crafted features. Images were segmented with colour thresholds, and colour, texture and shape descriptors such as colour co-occurrence matrices were extracted and classified with support vector machines or neural networks [8, 9]. Barbedo's survey of these techniques [9] notes that they work well under controlled conditions but are sensitive to illumination, background and the variability of symptoms, and that features must be engineered separately for each crop and disease.

The release of the PlantVillage dataset [10], containing tens of thousands of labelled leaf images photographed against uniform backgrounds, enabled end-to-end deep learning. Mohanty *et al.* [2] trained AlexNet and GoogLeNet on 38 classes and reported 99.35% accuracy on a held-out split. Ferentinos [3] reported 99.53% accuracy with VGG on a larger dataset of 58 classes, and Too *et al.* [4] compared fine-tuned architectures and obtained 99.75% with DenseNet. Geetharamani and Arun Pandian [11] designed a nine-layer CNN trained on an augmented version of PlantVillage; the non-augmented version of their dataset is used in this project.

2.2 The Laboratory-to-Field Gap

Near-perfect accuracy on PlantVillage does not transfer to photographs taken in the field. Mohanty *et al.* [2] already observed that accuracy dropped to 31.4% when their model was tested on images collected from other sources. Barbedo [12, 13] analysed the factors responsible: limited dataset variety, complex backgrounds, varying illumination and capture conditions, symptom variation between growth stages, and the co-occurrence of several disorders on the same leaf. To address this, Singh *et al.* [14] assembled PlantDoc, a dataset of field photographs covering 13 crops and 27 classes largely aligned with PlantVillage, and showed that models trained on laboratory images perform poorly on it while training on field images improves results considerably.

2.3 Lightweight Architectures and Transfer Learning

Residual networks [15] made very deep models trainable, but deployment on phones and low-cost devices favours efficient architectures. MobileNetV2 [16] introduced inverted residual blocks with linear bottlenecks; MobileNetV3 [5] combined them with squeeze-and-excitation modules, the hard-swish activation and architecture search to improve the accuracy–latency trade-off further. EfficientNet [17] scales depth, width and resolution jointly. All of these are usually initialised with weights pre-trained on ImageNet [18]; Yosinski *et al.* [19] showed that features learned on large generic datasets transfer well to new tasks, which makes it possible to

obtain strong results with comparatively little task-specific data and compute.

2.4 Reliability: Calibration and Selective Prediction

A diagnostic tool must indicate how much its output can be trusted. Guo *et al.* [20] showed that modern neural networks are poorly calibrated — their softmax confidence does not match their empirical accuracy — and that a single temperature parameter fitted on validation data largely corrects this. Selective classification [21] formalises the option to abstain: by rejecting inputs whose confidence falls below a threshold, the error rate on accepted inputs can be reduced at the cost of coverage. For crop diagnosis this trade-off is asymmetric, because recommending the wrong treatment is more harmful than asking for another photograph.

2.5 Comparison of Existing Approaches

Table 1 summarises the principal families of approaches together with their strengths and limitations.

Table 1: Comparison of approaches to image-based plant disease recognition

Approach	Representative work	Strengths	Limitations
Hand-crafted features with classical classifiers	[8, 9]	Interpretable; low compute; little training data	Sensitive to background and lighting; features engineered per crop; limited accuracy
Deep CNNs trained on laboratory images	[2, 3, 4, 11]	Very high accuracy on curated test sets; end-to-end learning	Accuracy collapses on field photographs; single label only; no severity or advice
Lightweight transfer-learning models	[16, 5, 17]	Fast inference; suitable for mobile and edge devices	Same domain-shift problem when trained on laboratory data
Field-image datasets	[14]	Realistic backgrounds and capture conditions	Small and noisy; lower absolute accuracy
This work (AgriVision AI)	—	Laboratory + field training; segmentation; severity; nutrient screening; context-aware advice; calibrated abstention; three interfaces	Field accuracy still moderate; nutrient and severity estimation rule-based

2.6 Research Gap

Most published systems report a single class label and its accuracy on a held-out split of the same laboratory dataset. Few of them estimate severity, distinguish nutrient deficiencies from disease, produce actionable and context-dependent recommendations, communicate calibrated confidence, abstain when unsure, or measure how performance changes on field photographs.

AgriVision AI is designed to close these gaps in one deployable system and to evaluate it rigorously on both laboratory and field images.

CHAPTER 3

System Design

3.1 Requirements

The functional and non-functional requirements derived from the problem statement are listed in Table 2.

Table 2: System requirements

ID	Requirement	Type
R1	Accept a leaf photograph by upload or camera capture	Functional
R2	Reject unusable images (no leaf, blurred, too dark or too bright) with guidance	Functional
R3	Identify crop and condition; route into disease, pest, nutrient deficiency or healthy	Functional
R4	Estimate severity and a 0–100 health score	Functional
R5	Provide treatment, prevention and context-aware recommendations	Functional
R6	Generate a downloadable crop-health report and keep a per-field history	Functional
R7	Always display confidence; abstain below a threshold	Reliability
R8	Diagnose an image in under 2 seconds on a laptop CPU; train without a GPU	Performance
R9	Expose the pipeline through an API so that new front ends need no model changes	Extensibility
R10	Operate without internet access once installed	Deployment

3.2 System Architecture

The system is organised in six layers, shown in Figure 1. Clients (the web application and the Android application) communicate with a REST service. The vision layer validates and pre-processes the photograph and segments the leaf. The decision layer classifies the image, applies the abstention rule, routes the result into a category, estimates severity and screens for nutrient deficiencies. The knowledge layer turns the diagnosis into recommendations, and the output layer produces the report and stores the diagnosis, which in turn feeds back into the recommendation engine to detect recurring outbreaks.

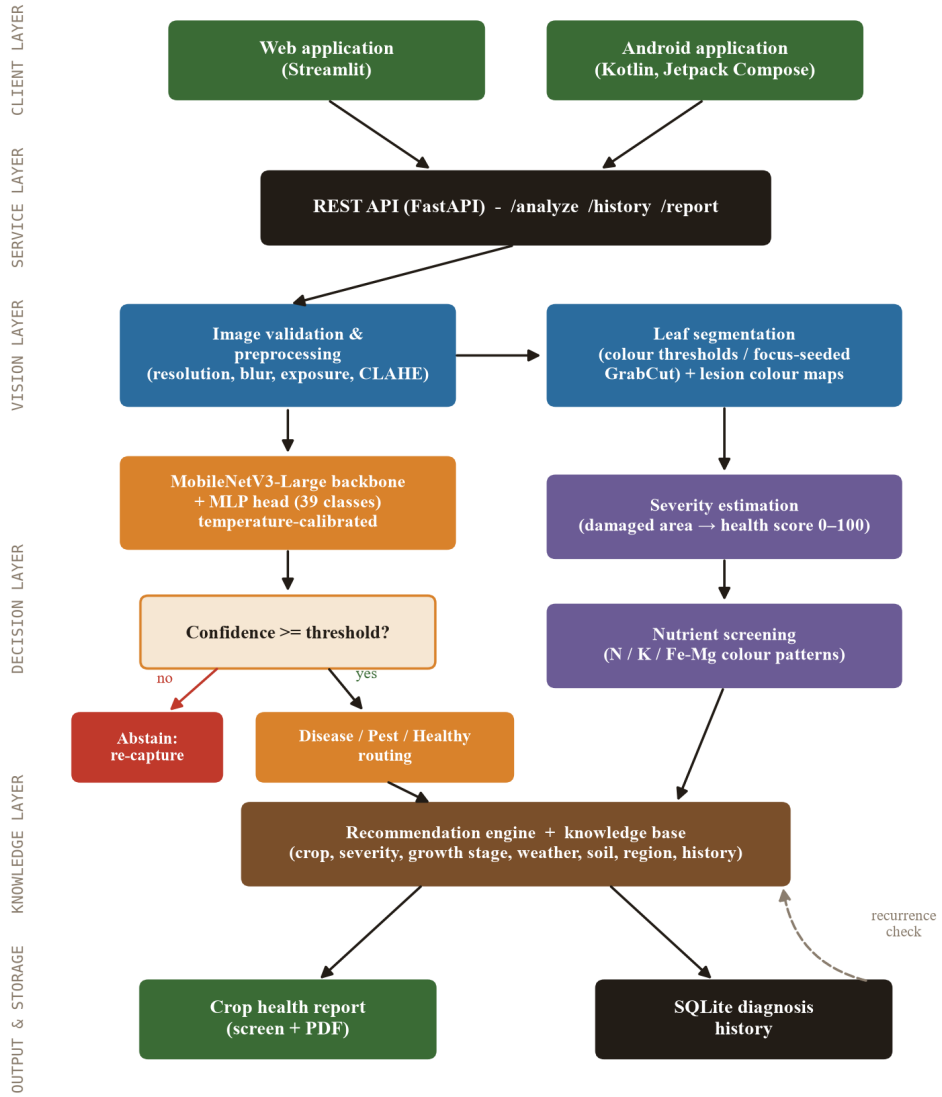


Figure 1: Layered architecture of AgriVision AI. Arrows show the flow of a single diagnosis; the dashed arrow shows the history lookup used to escalate recurring outbreaks

3.3 End-to-End Workflow

A single diagnosis proceeds through the following steps:

1. The user captures or uploads a photograph and optionally supplies the crop, field name, region, growth stage and sensor readings (temperature, humidity, soil moisture, soil pH).
2. The image is decoded with EXIF orientation applied, resized to at most 1024 pixels on its longer side and validated for resolution, blur, exposure and the presence of a leaf; dark images are enhanced.
3. The leaf is segmented, and each leaf pixel is labelled as healthy, chlorotic, necrotic or powdery.

4. The classifier produces calibrated probabilities over 39 classes; if the crop is supplied, probabilities are restricted to that crop.
5. If the most probable class is the background class, or confidence is below the applicable threshold, the system abstains and asks for a new photograph.
6. Otherwise the result is routed into a category, the health score and severity band are computed, and nutrient patterns are checked for leaves classified as healthy.
7. The recommendation engine combines the knowledge base with severity, context and field history.
8. The result is displayed, stored in the database and made available as a PDF report.

3.4 Technology Stack

Table 3: Technology stack

Layer	Technology	Purpose
Computer vision	OpenCV 5 [22], Pillow, NumPy	Decoding, validation, CLAHE, segmentation, colour analysis
Machine learning	PyTorch 2 [23], torchvision, scikit-learn [24]	MobileNetV3 backbone, head training, metrics
Backend	Python 3, FastAPI, Uvicorn	REST API and model serving
Web front end	Streamlit	Interactive web application
Mobile front end	Kotlin, Jetpack Compose, OkHttp	Native Android application (Android 9+)
Storage	SQLite	Diagnosis history and recurrence detection
Reporting	fpdf2, Matplotlib	PDF crop-health reports and evaluation plots

CHAPTER 4

Methodology

4.1 Datasets and Data Preparation

Two public datasets are used (Table 4). PlantVillage [10, 11] provides laboratory images of 38 crop conditions photographed on uniform backgrounds, together with a class of background images without leaves, which the system uses to reject photographs that do not contain a leaf. To keep training feasible on a CPU and to limit class imbalance, at most 400 images were sampled per class with a fixed random seed, giving 15,131 images that were split 70/15/15 into training, validation and test sets with stratification by class. Unreadable files were removed during preparation.

PlantDoc [14] contributes photographs taken in natural conditions, with foliage, soil, hands and structures in the background. Its class names were mapped onto the PlantVillage labels (for example, *Tomato Early blight leaf* to *Tomato – Early Blight*), which covers 28 of the 39 classes. PlantDoc's official test split was kept aside as an independent field test set; its training split was divided 85/15 into training and validation subsets.

Table 4: Datasets and splits used in this work

Dataset	Image type	Classes	Train	Validation	Test
PlantVillage	Laboratory, plain background	39	10,591	2,270	2,270
PlantDoc	Field, natural background	28 (mapped)	2,005	337	236
Total		39	**12,596**	**2,607**	**2,506**

4.2 Image Validation and Preprocessing

Photographs are first decoded with their EXIF orientation applied and downscaled so that the longer side is at most 1024 pixels. Four quality checks follow. The resolution must be at least 64 pixels on the shorter side. Sharpness is measured as the variance of the Laplacian of the grey-scale image resized to 512 × 512 pixels [25]:

$$B = \mathrm{Var}(\nabla^2 I) \tag{1}$$

Images with $B < 40$ are reported as blurred. Mean brightness must lie between 40 and 225, and the segmented leaf must cover at least 5% of the frame. Failed checks are shown to the user as specific guidance, for example to photograph in daylight or to fill the frame with the leaf. Images with mean brightness below 70 are enhanced with contrast-limited adaptive histogram equalisation (CLAHE) [26] applied to the lightness channel of the Lab colour space (clip limit

2.0, 8 × 8 tiles). For classification, the image is resized to 224 × 224 pixels and normalised channel-wise with the ImageNet statistics:

$$\hat{x}_c = \frac{x_c/255 - \mu_c}{\sigma_c}, \quad \mu = (0.485, 0.456, 0.406), \ \sigma = (0.229, 0.224, 0.225) \tag{2}$$

4.3 Leaf Segmentation

Segmentation serves two purposes: it measures how much of the leaf is damaged, and it provides the visual explanation shown to the user. Two strategies are used, selected automatically from the image border.

Plain-background photographs

Plant tissue is saturated in colour, while backgrounds such as paper, benches or walls are not. A pixel is marked as plant if its saturation exceeds 45 and its value exceeds 50 with a hue in the red–yellow–green range, or if its excess-green index [27] exceeds 25:

$$\mathrm{ExG} = 2G - R - B \tag{3}$$

Coloured backgrounds are removed by estimating the background chroma from a border band covering 4% of the image. If the border is uniform (70% of border pixels lie within 6 units of the median in the Lab a*b* plane), every pixel whose chroma is within 7 units of that median is treated as background:

$$d(p) = \left\| (a_p^*, b_p^*) - (\tilde{a}^*, \tilde{b}^*) \right\|_2 < 7 \tag{4}$$

Because only chroma is compared, the background's own shadows are removed as well. The mask is cleaned with morphological opening and closing, the largest connected component is kept, and its holes are filled so that lesions inside the leaf are included.

Field photographs

When the border is not uniform, colour cannot separate the leaf from the foliage behind it. The method developed here exploits the camera's depth of field: the photographed leaf is in focus, while the background is blurred. A local focus map is computed from the absolute Laplacian, smoothed with a Gaussian of standard deviation 3% of the image size and normalised by its 99th percentile:

$$F(p) = \frac{(G_\sigma * |\nabla^2 I|)(p)}{P_{99}} \tag{5}$$

These cues seed the GrabCut algorithm [28], which models foreground and background with Gaussian mixture models and refines the boundary with graph cuts. Pixels with $F < 0.12$ are marked as definite background, in-focus plant-coloured pixels ($F > 0.35$) as probable foreground, and all others as probable background. Five iterations are run on a copy downscaled to 400 pixels, and the component that best combines size and centrality is kept. Figure 2 compares the

two strategies on a field photograph: colour thresholding selects 87% of the frame, including the fence and post, whereas the focus-seeded GrabCut isolates the leaf.

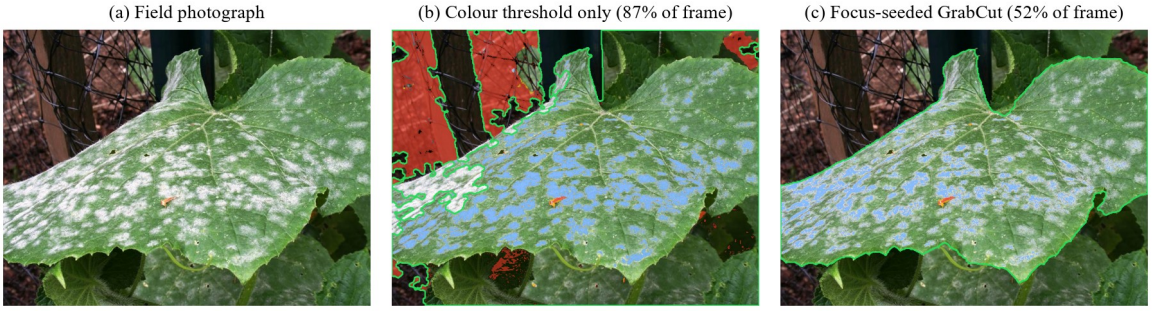


Figure 2: Leaf segmentation on a field photograph from the PlantDoc test set. Green outline: detected leaf; blue: powdery growth; red: tissue classified as necrotic. Colour thresholding alone (b) includes the background, while focus-seeded GrabCut (c) isolates the leaf

Tissue classification

Within the leaf mask, each pixel is labelled in HSV space (OpenCV hue scale 0–179) as *powdery* (saturation below max(15, 0.35·median saturation) and value above max(170, median value + 60)), *healthy green* (hue 28–95, saturation above 20, value above 25), *chlorotic* (hue 15–28, saturation above 60, value above 90) or *necrotic* (remaining pixels with saturation above 35 or value below 70). Pixels matching none of these, such as glare and deep shadow, are excluded from the percentages. The adaptive thresholds for powdery tissue avoid confusing leaf veins and glossy highlights with mildew.

4.4 Classification Model

Architecture

The classifier uses MobileNetV3-Large [5] pre-trained on ImageNet (torchvision weights IMAGENET1K_V2) as a feature extractor. The original 1000-class output layer is removed; the network up to the 1280-dimensional penultimate layer (convolutional features, global average pooling, a fully connected layer and hard-swish activation) maps an image x to an embedding:

$$z = f_\theta(x) \in \mathbb{R}^{1280} \tag{6}$$

A classification head with one hidden layer of 512 units maps the embedding to logits for the 39 classes:

$$h(z) = W_2\,\mathrm{ReLU}(W_1 z + b_1) + b_2, \quad W_1 \in \mathbb{R}^{512\times 1280},\ W_2 \in \mathbb{R}^{39\times 512} \tag{7}$$

Dropout with probability 0.3 is applied before each linear layer. The head has 675,879 trainable parameters, compared with 4,226,478 parameters in the frozen backbone. A single linear layer was evaluated first; the hidden layer improved both laboratory and field validation accuracy and was retained.

Training by feature extraction

Fine-tuning the entire network on a CPU would require a forward and backward pass through the backbone for every image in every epoch. Because the backbone is frozen, each image needs to pass through it only once: the embeddings of all images are computed and cached, after which the head is trained on the cached embeddings in seconds per epoch [19]. Field photographs are mirrored horizontally as augmentation and oversampled four times, so that the 2,005 field images carry weight comparable to the 10,591 laboratory images; the final training set contains 26,631 embeddings. The loss is cross-entropy with label smoothing [29]:

$$\mathcal{L} = -\sum_{k=1}^{K} q_k \log p_k, \qquad q_k = (1-\varepsilon)\, y_k + \frac{\varepsilon}{K}, \quad \varepsilon = 0.05 \tag{8}$$

Optimisation uses AdamW [30] with a cosine learning-rate schedule [31]. The model is selected by the mean of laboratory and field validation accuracy, with early stopping after 15 epochs without improvement. Table 5 lists the hyperparameters.

Table 5: Model and training hyperparameters

Parameter	Value
Backbone	MobileNetV3-Large, ImageNet-1K (V2) weights, frozen
Input size / embedding size	224 × 224 × 3 / 1280
Classification head	1280 → 512 → 39, ReLU, dropout 0.3
Optimiser	AdamW, learning rate 10^{-3}, weight decay 10^{-4}
Batch size / maximum epochs	128 / 80 (cosine schedule)
Early stopping	patience 15; stopped after 52 epochs, best epoch 37
Label smoothing	0.05
Field-image weighting	horizontal mirroring, 4× oversampling
Training embeddings	26,631
Temperature T	0.725
Abstention thresholds	0.60 (plain background), 0.73 (field)

Confidence calibration

Raw softmax probabilities are not reliable confidence estimates [20]. After training, a temperature T is fitted on the combined validation set by minimising the negative log-likelihood, and the probability of class k is computed as:

$$p_k = \frac{\exp\left(h_k(z)/T\right)}{\sum_j \exp\left(h_j(z)/T\right)} \tag{9}$$

The fitted value $T = 0.725$ is below one, which sharpens the distribution. This is expected: label smoothing deliberately trains the model to be less confident, and temperature scaling restores probabilities that match observed accuracy.

4.5 Abstention and the False Diagnosis Rate

A wrong treatment recommendation is more harmful than a request for another photograph, so accuracy alone is not the most relevant metric. Following the selective classification framework [21], for a confidence threshold τ the system answers only images whose top probability c is at least τ. Two quantities characterise the behaviour, coverage and the false diagnosis rate (FDR):

$$\mathrm{Cov}(\tau) = \frac{|\{i : c_i \geq \tau\}|}{N}, \qquad \mathrm{FDR}(\tau) = \frac{|\{i : c_i \geq \tau,\ \hat{y}_i \neq y_i\}|}{|\{i : c_i \geq \tau\}|} \tag{10}$$

For plain-background images the threshold is fixed at 0.60. Field photographs are much harder, so they receive their own threshold, selected automatically during training as the smallest value that keeps the false diagnosis rate on the field validation set at or below 20%:

$$\tau_f = \min \{\tau \geq 0.60 \ :\ \mathrm{FDR}_{\mathrm{val}}(\tau) \leq 0.20\} \tag{11}$$

This yields $\tau_f = 0.73$. The photograph type is determined by the same border-uniformity test used for segmentation. When the system abstains, it shows the three most probable classes and specific advice for re-capture instead of a treatment.

4.6 Category Routing and Nutrient Screening

Each of the 39 classes belongs to one category in the knowledge base: disease (25 classes), pest (1 class, spider-mite damage), healthy (12 classes, one per crop) or invalid (1 class, no leaf). Category probabilities are obtained by summing class probabilities:

$$P(\mathrm{cat}) = \sum_{k \in \mathrm{cat}} p_k \tag{12}$$

Nutrient deficiencies are not labelled in either dataset, so they are screened with explainable colour rules based on well-known deficiency signatures [32] (Table 6). The rules are evaluated on the leaf mask: the margin band is the outer 6% of the leaf's equivalent diameter, and vein-like green structures are identified as green pixels removed by a morphological opening with a kernel of 1.5% of the diameter. A deficiency is reported as the primary diagnosis only if the classifier considers the leaf healthy and the stress index reaches 0.25; its probability is taken as the healthy probability multiplied by the stress index. Results are always presented as probable, with a recommendation to confirm by soil or tissue testing before fertiliser is applied.

Table 6: Nutrient-deficiency signatures and screening rules

Deficiency	Visual signature	Rule applied to the segmented leaf
Nitrogen (N)	Uniform yellowing; older leaves first; reduced growth	Chlorotic fraction ≥ 25% without vein pattern or margin concentration
Potassium (K)	Yellow/brown scorching of leaf margins	Margin damage > 30% and exceeds interior damage by > 20 percentage points
Iron / Magnesium (Fe, Mg)	Interveinal chlorosis: yellow lamina, green veins	Chlorotic fraction ≥ 12% and > 35% of green pixels lie in thin vein-like lines

4.7 Severity Estimation

The damaged fraction a of the leaf is the share of classified leaf pixels that are chlorotic (Y), necrotic (B) or powdery (W) rather than healthy green (G):

$$a = \frac{|Y| + |B| + |W|}{|G| + |Y| + |B| + |W|} \tag{13}$$

For leaves diagnosed with a disease or pest, the health score is a non-linear function of the damaged fraction, so that the first lesions reduce the score less than the loss of the remaining healthy tissue:

$$S = 100\,(1 - a)^{1.6} \tag{14}$$

Damage of 5%, 15%, 30% and 45% gives scores of 92, 77, 56 and 38 respectively (Figure 3). Because a confirmed disease is never healthy, the score of a diseased leaf is capped at 89. Aggressive or systemic conditions carry a minimum severity regardless of visible area: late blight, Tomato Yellow Leaf Curl Virus and Tomato Mosaic Virus are at least *Moderate*, and citrus greening (Huanglongbing) is always *Severe*. Healthy leaves score between 90 and 100 depending on confidence and residual discolouration. Table 7 lists the bands.

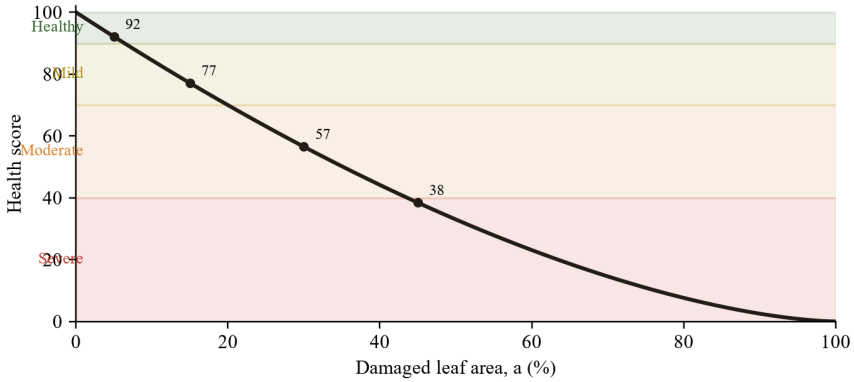


Figure 3: Mapping from damaged leaf area to the crop health score, with the four severity bands

Table 7: Severity bands and associated urgency

Health score	Severity band	Urgency of the recommendation
90 – 100	Healthy	Routine weekly monitoring
70 – 89	Mild	Monitor closely; re-check in 3–4 days; treat if spreading
40 – 69	Moderate	Act this week; apply treatment
0 – 39	Severe	Act today; isolate affected plants; inspect the whole field

4.8 Recommendation Engine

A knowledge base stores, for each of the 39 classes, the crop, condition name, category, cause type (fungal, bacterial, viral, oomycete or pest), causal organism, typical symptoms, cultural actions, a treatment option, prevention measures, an optional minimum severity and the environmental conditions that favour the pathogen. Treatments are described generically (for example, a locally approved protectant fungicide such as chlorothalonil or mancozeb) and always point the user to the local agricultural extension centre (Krishi Vigyan Kendra) for approved products and doses. The engine combines the knowledge base entry with the factors in Table 8.

Table 8: Context factors used by the recommendation engine

Factor	Effect on the recommendation
Severity	Selects the action set and urgency (Table 7); treatment is included from *Moderate* upwards
Temperature and humidity	Compared with the pathogen's favourable range; if conditions favour spread, a warning is added and treatment is recommended even for mild cases
Growth stage	Seedling: remove rather than treat; flowering: spray outside pollinator activity; fruiting: observe the pre-harvest interval
Soil moisture and pH	Low moisture supports spider-mite stress advice; high pH supports iron deficiency, low pH magnesium deficiency
Diagnosis history	The same problem in the same field within 30 days is reported as a recurring outbreak and escalated to expert consultation, resistant varieties and crop rotation
Region	Reminder to use only products approved in the user's state
Crop selection	If the user specifies the crop, only that crop's classes compete in the classifier

CHAPTER 5

Implementation

5.1 Software Organisation

The core logic is implemented as a Python package, `agrivision`, that is shared by all interfaces, so that the web application, REST API and Android application produce identical results. Table 9 lists its modules.

Table 9: Modules of the agrivision package

Module	Responsibility
preprocessing.py	Decoding, validation, CLAHE, both segmentation strategies, tissue colour maps, overlays
model.py	MobileNetV3 backbone, classification head, calibrated inference
pipeline.py	End-to-end workflow, abstention, category routing, result assembly
severity.py	Damaged-area statistics, health score and severity band
nutrient.py	Nutrient-deficiency screening
recommender.py, knowledge_base.json	Recommendation engine and knowledge base
database.py	SQLite history, recurrence queries
report.py	PDF crop-health reports
scripts/	Data preparation, feature extraction and training, evaluation, automated tests

5.2 Web Application

The web application, built with Streamlit, follows a single-screen design: capture or upload, analyse, read, and download the report. Optional field context is placed in a collapsed panel so that it does not distract users who do not need it. Figure 4 shows the input screen and Figure 5 the result for a maize leaf: the category, diagnosis, cause, confidence and severity, the health score bar, observed symptoms and prioritised actions. Confidence is colour-coded, and a low-confidence result looks visibly different from a confident one. An expandable panel (Figure 6) explains how the result was reached: the segmentation, the damage map, category probabilities, the five most probable classes, tissue percentages, nutrient screening and image-quality measurements. Further tabs show the history of each field with a health-score trend, and the model's evaluation results.

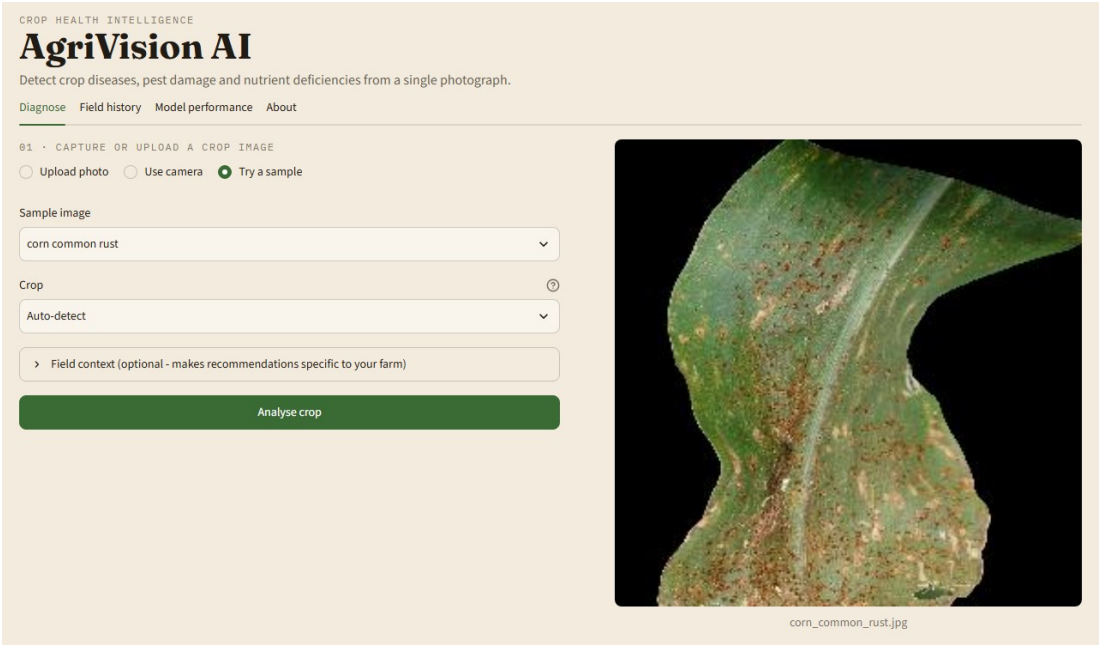


Figure 4: Web application: input screen with a sample image selected

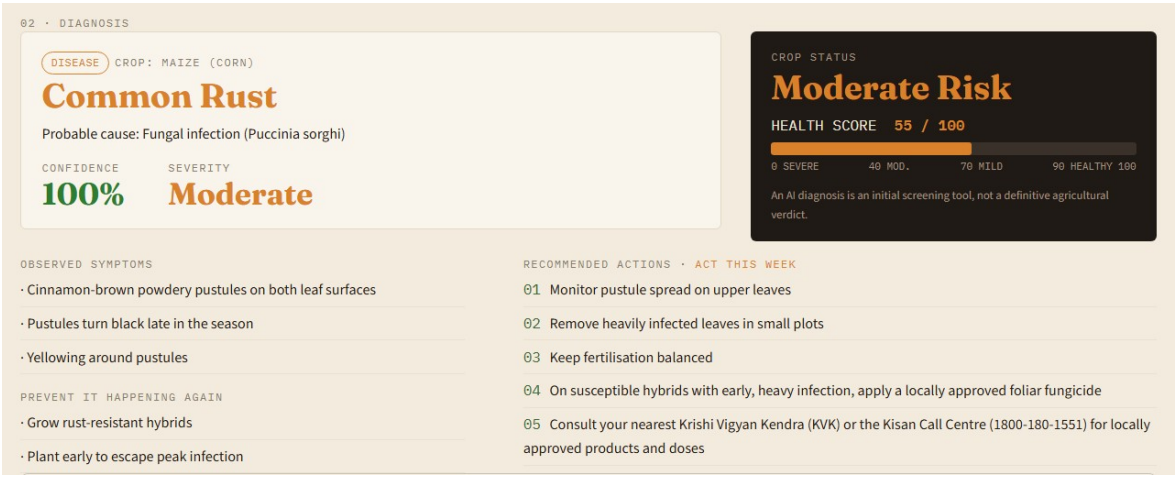


Figure 5: Web application: diagnosis, crop status and recommended actions for a maize leaf with common rust

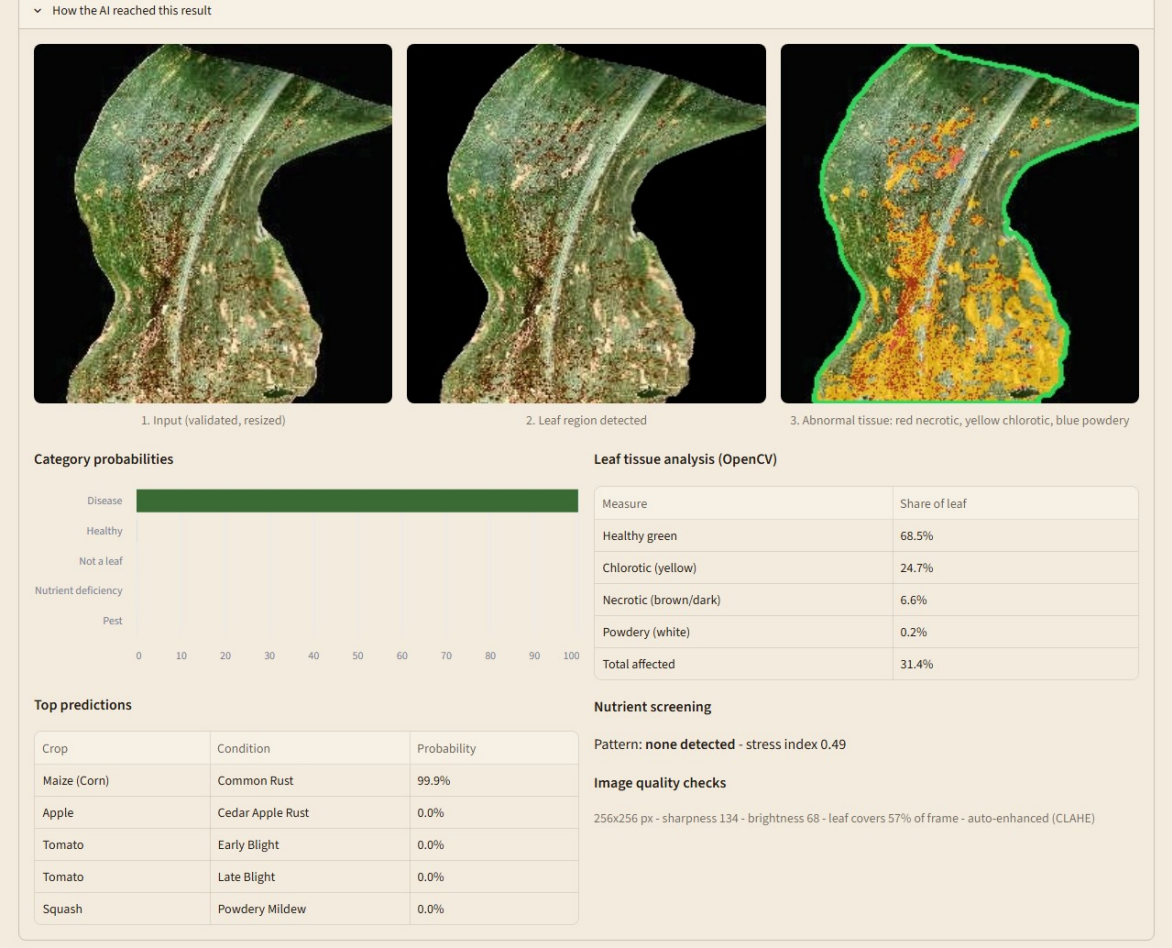


Figure 6: Web application: explanation panel with segmentation, damage map, category probabilities, top predictions and tissue analysis

5.3 REST API

The FastAPI service exposes the pipeline to any client (Table 10). The `/analyze` endpoint accepts the image as multipart form data together with the optional context fields and returns the full result as JSON; on request it also returns the original image, the segmented leaf and the damage map as base64-encoded JPEG images. Interactive documentation is generated automatically.

Table 10: REST API endpoints

Method	Endpoint	Description
POST	/analyze	Diagnose an image; optional crop, field, region, growth stage, temperature, humidity, soil moisture, soil pH, include_images
GET	/history	List diagnoses, optionally filtered by field
GET	/diagnoses/{id}	Retrieve one stored diagnosis
GET	/report/{id}	Download the PDF crop-health report
GET	/crops	List supported crops
GET	/health	Service and model status

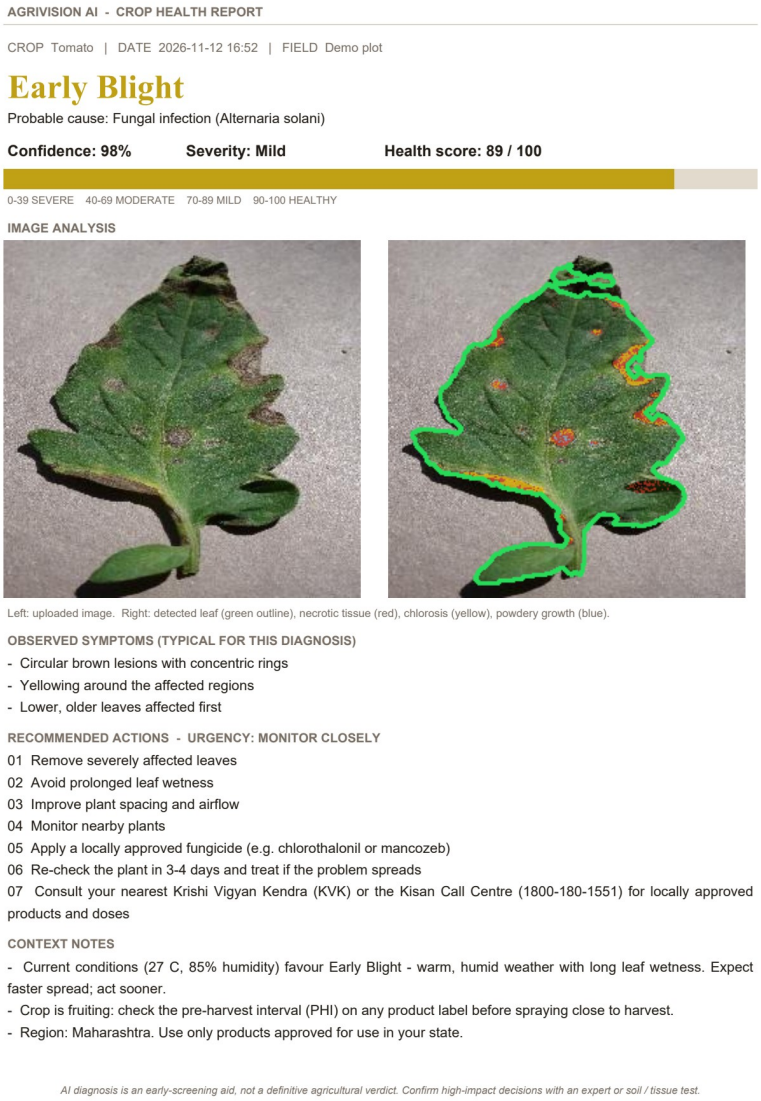
5.4 Android Application

The Android application is written in Kotlin with Jetpack Compose and supports Android 9 and later. It contains four screens: *Diagnose* (camera capture or gallery selection, crop selection and field context), *Result* (the same information as the web application, including the damage map), *History* (previous diagnoses with health scores) and *Settings* (server address and connection test). Photographs are decoded with their orientation applied, downscaled to 1600 pixels and compressed before upload, which keeps transfers fast on mobile networks. PDF reports are downloaded and opened in the phone's PDF viewer or shared through other applications.

The application connects to the API over a local network. In the intended field setup the laptop running the server joins the phone's mobile hotspot, which requires no internet connection; a launcher script on the laptop prints the address to enter in the application. The application is distributed as a 23 MB APK.

5.5 Crop Health Report

Every diagnosis can be exported as a PDF report (Figure 7). The report contains the crop, date and field, the diagnosis and probable cause, confidence, severity, the health score bar with its scale, the input image and damage map, observed symptoms, numbered recommended actions with their urgency, context notes, prevention measures, the model output and a disclaimer that the diagnosis is an early-screening aid.



AGRIVISION AI - CROP HEALTH REPORT

CROP Tomato | DATE 2026-11-12 16:52 | FIELD Demo plot

Early Blight

Probable cause: Fungal infection (Alternaria solani)

Confidence: 98% **Severity: Mild** **Health score: 89 / 100**

0-39 SEVERE 40-69 MODERATE 70-89 MILD 90-100 HEALTHY

IMAGE ANALYSIS

Left: uploaded image. Right: detected leaf (green outline), necrotic tissue (red), chlorosis (yellow), powdery growth (blue).

OBSERVED SYMPTOMS (TYPICAL FOR THIS DIAGNOSIS)

- Circular brown lesions with concentric rings
- Yellowing around the affected regions
- Lower, older leaves affected first

RECOMMENDED ACTIONS - URGENCY: MONITOR CLOSELY

01 Remove severely affected leaves
02 Avoid prolonged leaf wetness
03 Improve plant spacing and airflow
04 Monitor nearby plants
05 Apply a locally approved fungicide (e.g. chlorothalonil or mancozeb)
06 Re-check the plant in 3-4 days and treat if the problem spreads
07 Consult your nearest Krishi Vigyan Kendra (KVK) or the Kisan Call Centre (1800-180-1551) for locally approved products and doses

CONTEXT NOTES

- Current conditions (27 C, 85% humidity) favour Early Blight - warm, humid weather with long leaf wetness. Expect faster spread; act sooner.
- Crop is fruiting: check the pre-harvest interval (PHI) on any product label before spraying close to harvest.
- Region: Maharashtra. Use only products approved for use in your state.

AI diagnosis is an early-screening aid, not a definitive agricultural verdict. Confirm high-impact decisions with an expert or soil / tissue test.

Figure 7: First page of a generated crop-health report for a tomato leaf with early blight, with field context (fruiting stage, 27 °C, 85% humidity)

5.6 Testing

Two automated test suites verify the system end to end. The first runs every sample image through the pipeline, generates a PDF report for each and exercises all REST endpoints, including report download and history. The second drives the Streamlit interface headlessly: it selects samples, submits field context and sensor readings, checks the rendered verdicts, including rejection of a non-leaf image, abstention on an ambiguous image and its resolution when the crop is specified, and confirms that every tab renders without errors. Both suites pass on the final system.

CHAPTER 6

Results and Analysis

6.1 Experimental Setup

All experiments were run on a laptop with an Intel Core i5-1035G1 processor and 8 GB of memory, without a GPU. The laboratory test set consists of 2,270 PlantVillage images not used in training or model selection; the field test set consists of the 236 images of PlantDoc's official test split. Accuracy, macro-averaged precision, recall and F1-score, category routing accuracy (whether the disease / pest / healthy / no-leaf category is correct), coverage and false diagnosis rate are reported.

6.2 Performance on Laboratory Images

Table 11 summarises the results on the laboratory test set. The classifier reaches 97.0% accuracy with balanced precision and recall, and assigns the correct category to 98.9% of images. Without abstention, 3.0% of images would be misdiagnosed; at the 0.60 threshold, 97.2% of images are answered and the false diagnosis rate falls to 1.81%.

Table 11: Results on the laboratory test set (2,270 images, 39 classes)

Metric	Value
Accuracy	**97.0%**
Macro precision / recall / F1-score	0.970 / 0.970 / 0.970
Category routing accuracy	98.9%
False diagnosis rate without abstention	3.0%
False diagnosis rate at threshold 0.60	**1.81%**
Coverage at threshold 0.60	97.2%

The confusion matrix (Figure 8) is strongly diagonal. The remaining errors are concentrated among visually similar diseases of the same crop: the lowest F1-scores are obtained for Tomato Early Blight (0.860), Maize Northern Leaf Blight (0.874), Maize Gray Leaf Spot (0.876), Tomato Spider Mite Damage (0.894) and Tomato Target Spot (0.903). Early blight and target spot both produce brown lesions with concentric rings, and northern leaf blight and gray leaf spot of maize both form elongated grey-tan lesions that differ mainly in shape and size, so these confusions mirror those of human observers. Per-class results are listed in Appendix A.

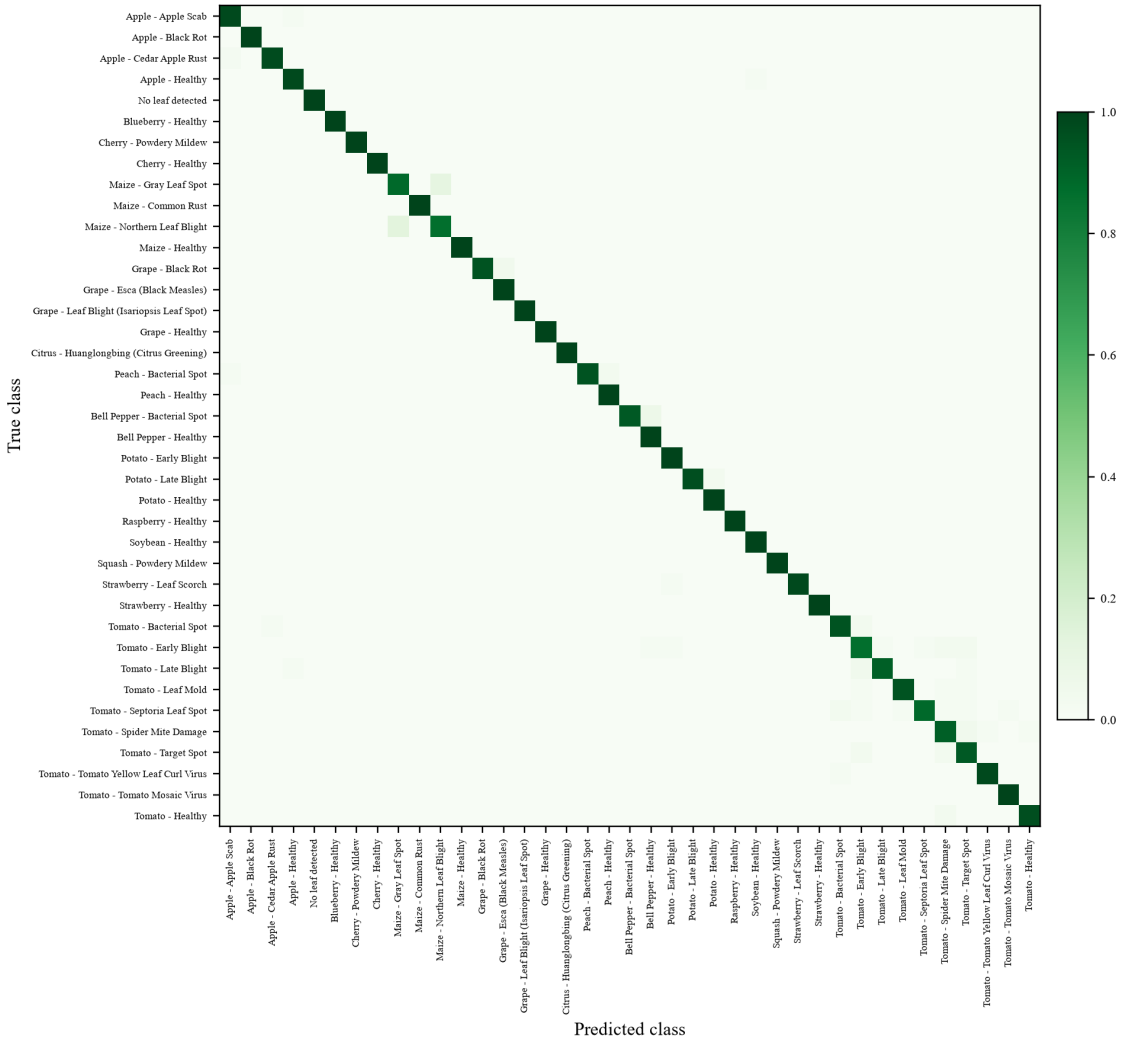


Figure 8: Row-normalised confusion matrix on the laboratory test set

6.3 Performance on Field Photographs

Field photographs are substantially harder (Table 12, Figure 9). A model trained only on laboratory images, with the same architecture, reaches 97.4% on the laboratory test set but only 19.9% on the field test set: it has learned to rely on cues, such as the uniform grey background, that are absent in the field. Adding the PlantDoc training images raises field accuracy to 53.8%, a factor of 2.7, while laboratory accuracy changes by only -0.5 percentage points. The correct class is among the three most probable in 73.7% of field images, and the category is correct in 89.8%, which matters because the category determines the type of advice.

Table 12: Laboratory versus field performance (236 field images)

Metric	Laboratory-only model	Final model
Accuracy on laboratory test set	97.4%	97.0%
Accuracy on field test set	19.9%	**53.8%**
Field top-3 accuracy	—	73.7%
Field macro F1-score	—	0.527
Field category routing accuracy	—	89.8%
Field coverage at threshold 0.73	—	55.1%
Field false diagnosis rate at threshold 0.73	—	27.7%

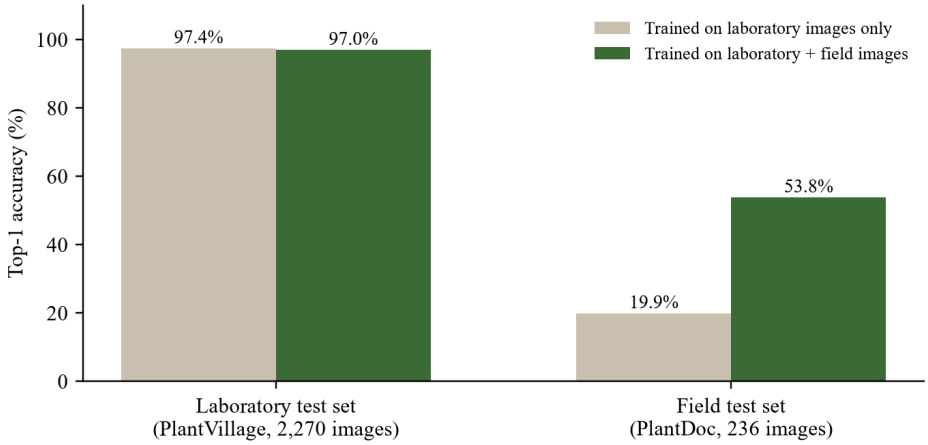


Figure 9: Effect of including field photographs in training on laboratory and field accuracy

Performance varies by crop (Figure 10). Grape, squash, raspberry and strawberry, whose field images are visually distinctive, reach high field accuracy, while tomato and potato, which have the largest number of similar-looking diseases, are among the most difficult. The per-crop field figures are based on small samples of 4 to 12 images per class and should be read as indicative.

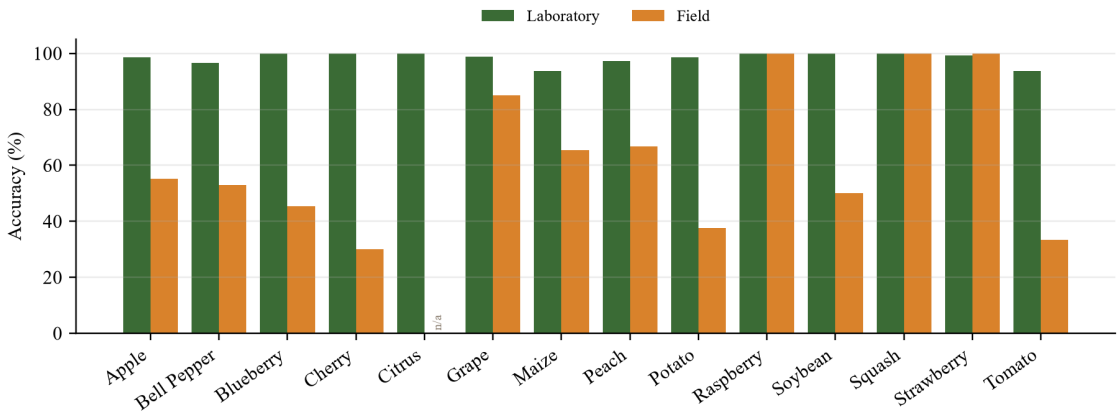


Figure 10: Accuracy by crop on the laboratory and field test sets. Citrus has no field test images

6.4 Abstention Trade-off

Figure 11 shows coverage and false diagnosis rate as functions of the confidence threshold. On laboratory images, almost all predictions are confident and correct, so a threshold of 0.60 removes roughly 40% of the remaining errors while answering 97.2% of images. On field photographs, the stricter threshold of 0.73 reduces the false diagnosis rate from 46.2% without abstention to 27.7%, at the cost of asking for a new photograph for 44.9% of field images. The threshold can be moved along this curve depending on how cautious a deployment needs to be.

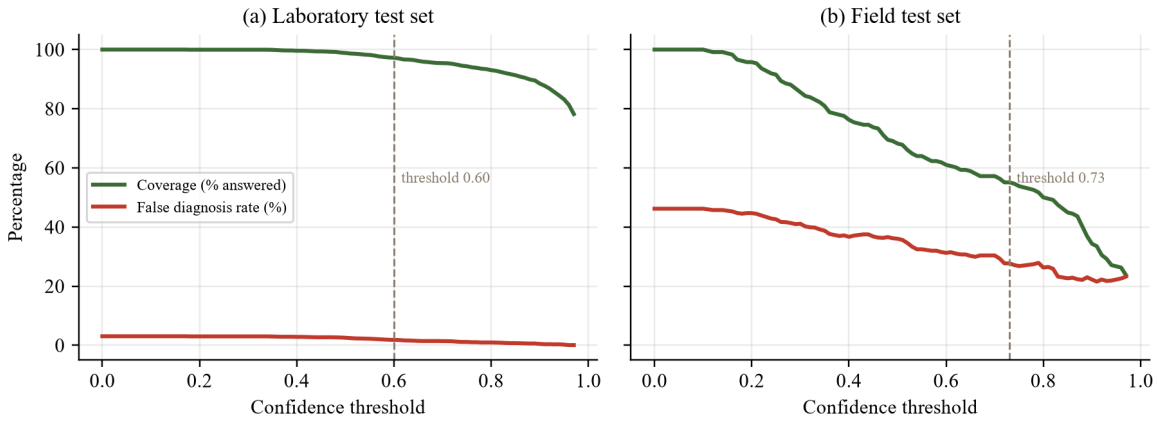


Figure 11: Coverage and false diagnosis rate as functions of the confidence threshold on the laboratory (a) and field (b) test sets; dashed lines mark the thresholds used

6.5 Training Behaviour

Figure 12 shows the validation accuracy during head training. Laboratory accuracy exceeds 94% within a few epochs, reflecting the quality of the pre-trained features, while field validation accuracy fluctuates around 60% on the small field validation set. The model from epoch 37, which maximises the mean of the two, was retained; training stopped after 52 epochs.

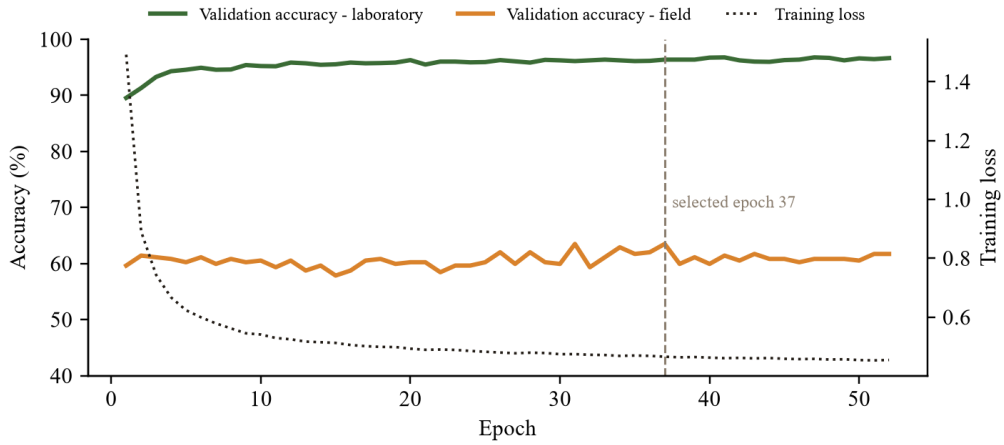


Figure 12: Validation accuracy on laboratory and field images and training loss during head training

6.6 Qualitative Results

Figure 13 shows the complete pipeline on two laboratory images and two field photographs. The laboratory tomato leaf with early blight is diagnosed with 98% confidence and 7% damaged tissue, giving a *Mild* rating, whereas the heavily affected potato leaf has 69% damaged tissue and is rated *Severe* with a health score of 16. The field photographs of cedar apple rust and grape black rot are correctly isolated from foliage and soil, and their lesions are highlighted in the damage map.

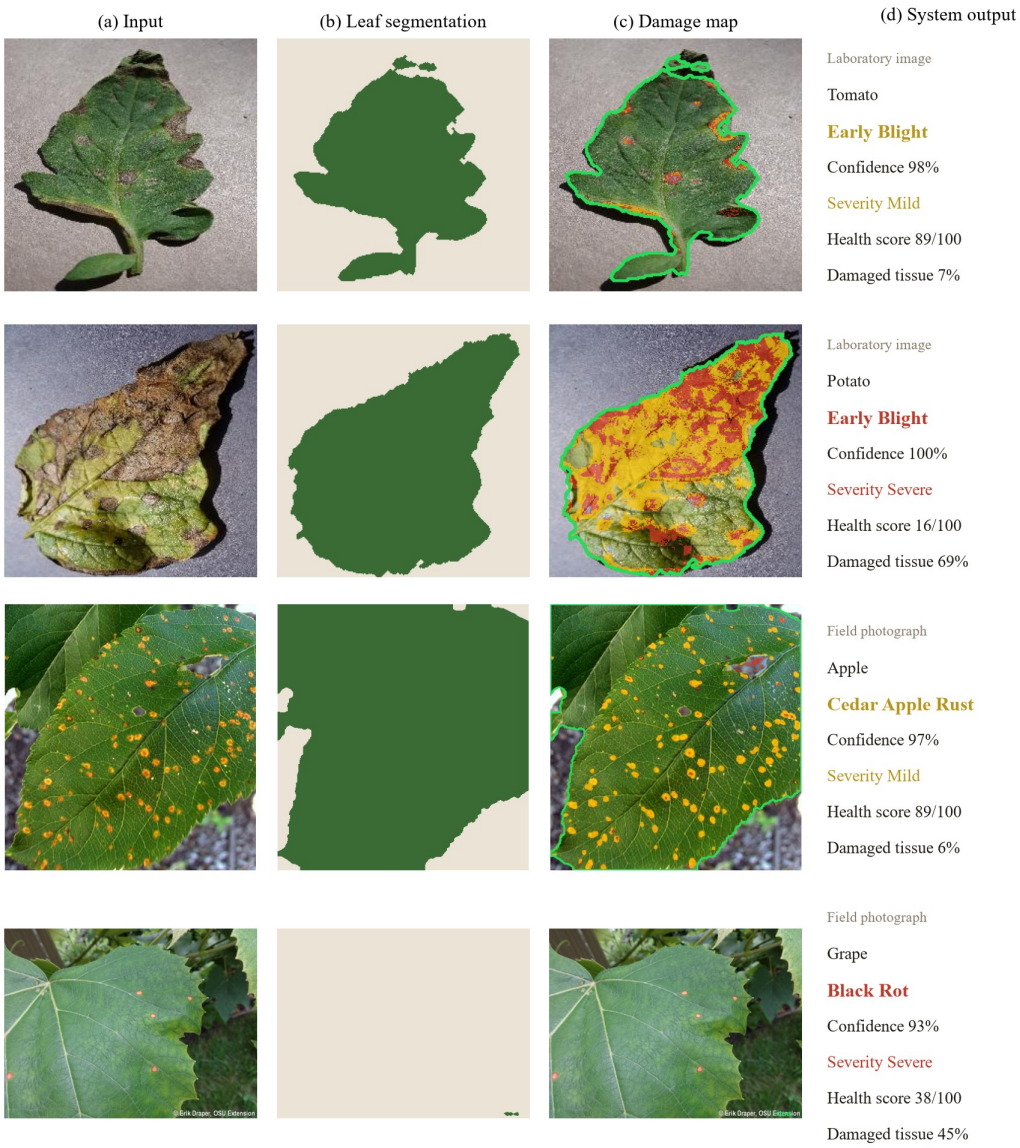


Figure 13: Qualitative results: input, leaf segmentation, damage map (red: necrotic, yellow: chlorotic, blue: powdery) and system output for two laboratory images (rows 1–2) and two field photographs (rows 3–4)

Figure 14 shows three characteristic difficulties on field photographs. In (a), the correct class is the most probable but with only 28% confidence, and the system correctly asks for a new

photograph instead of guessing. In (b), Septoria leaf spot is confidently classified as early blight; both diseases produce dark lesions surrounded by yellowing, and the photograph shows several overlapping leaves. In (c), the diagnosis of late blight is correct, but the dark, low-saturation lesion at the leaf edge is excluded by the segmentation, so the damaged area is underestimated; the minimum severity for late blight still yields a *Moderate* rating.

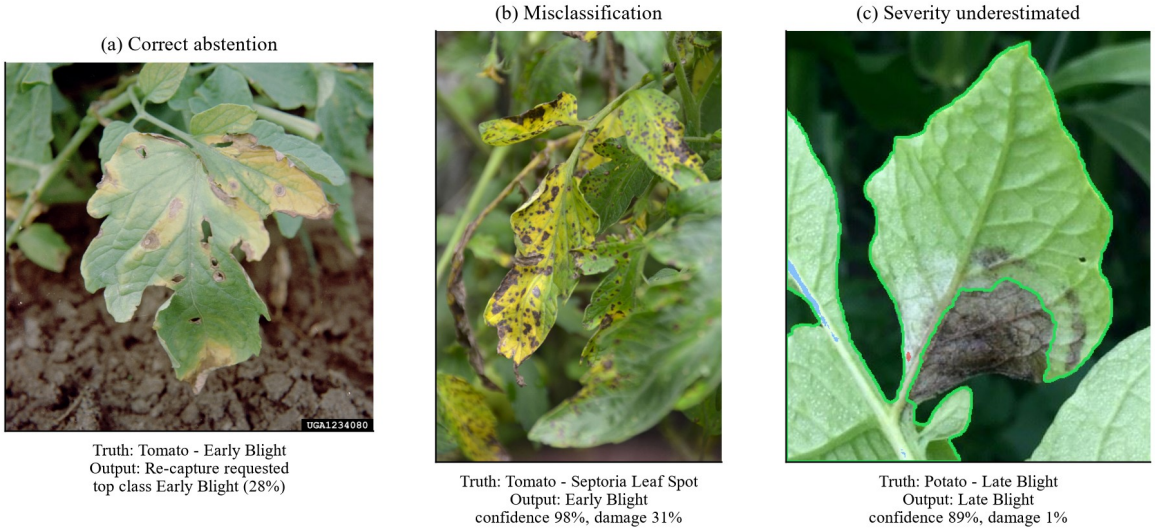


Figure 14: Challenging field photographs: correct abstention (a), misclassification between similar diseases (b) and underestimated damage area (c)

6.7 Computational Performance

Table 13 summarises the computational cost. All figures were measured on the laptop described above.

Table 13: Computational cost on a laptop CPU (Intel Core i5-1035G1, 8 GB RAM)

Item	Value
Feature extraction, all training/validation/test images (one-time)	≈ 10 minutes
Classification head training	≈ 3 minutes
Full diagnosis, plain-background photograph	0.06 s
Full diagnosis, field photograph (including GrabCut)	1.04 s
Model file size (backbone + head)	19.7 MB
Android application package	23 MB

CHAPTER 7

Discussion

7.1 Laboratory Accuracy Overstates Real-World Performance

The most important finding of this work is the size of the gap between laboratory and field performance. A model that is 97.4% accurate on PlantVillage is correct on only 19.9% of field photographs. This agrees with earlier observations [2, 12, 14] and shows that accuracy on a held-out split of a laboratory dataset should not be presented as an estimate of usefulness in the field. Including even a modest number of field photographs in training more than doubles field accuracy without harming laboratory accuracy, which indicates that data diversity, rather than model capacity, is the limiting factor. Field accuracy of 53.8% is, however, far from sufficient for unsupervised use, which motivates the design choices discussed next.

7.2 The Value of Calibration and Abstention

Because field accuracy is moderate, the system's behaviour on uncertain inputs is as important as its accuracy. Temperature scaling makes the reported confidence meaningful, and the abstention rule converts low-confidence predictions into requests for a better photograph, which costs the user a few seconds rather than a wrong treatment. Using a separate, stricter threshold for field photographs reflects the different difficulty of the two conditions; the photograph type is detected automatically, so the user does not need to make this distinction. The remaining false diagnoses on field images (Figure 14b) are mostly confusions between diseases of the same crop that share symptoms; in such cases the category, and therefore the broad type of response (for example, a fungal disease requiring sanitation and a fungicide), is usually still correct.

7.3 Design Trade-offs

Freezing the backbone and training only the head made it possible to train the complete model on a CPU in about a quarter of an hour, at the cost of not adapting the convolutional features to plant imagery. Severity and nutrient screening are implemented with explicit colour rules rather than learned models. This choice makes their output explainable — the damage map shows exactly which pixels were counted — and requires no labelled severity or deficiency data, which is not available in the public datasets; the price is lower robustness to unusual lighting and lesion colours. Finally, separating the pipeline into a shared package behind a REST API allowed three interfaces to be built without duplicating any diagnostic logic.

CHAPTER 8

Limitations and Future Work

8.1 Limitations

- **Field accuracy.** Accuracy on field photographs (53.8%) remains far below laboratory accuracy. The PlantDoc training set is small (2,005 images) and contains label noise and images with several leaves or plants.
- **Pest coverage.** The only pest class available in the training data is spider-mite damage; other pests such as aphids, caterpillars and thrips are not recognised.
- **Nutrient deficiencies** are screened with colour rules rather than a trained model and cannot be confirmed visually; soil or tissue testing is required before any fertiliser decision.
- **Severity estimation** depends on colour segmentation. On field photographs, dark low-saturation lesions at the leaf edge can be excluded from the leaf mask (Figure 14c), and early lesions that remain green-hued are not counted, so severity may be underestimated.
- **Single-leaf input.** The system analyses one leaf per photograph and does not assess whole plants, fruit or stems.
- **Connectivity.** The Android application requires a reachable server on the local network; inference does not yet run on the phone itself.
- **Recommendations** are generic within each class and must be adapted to locally approved products and regulations.

8.2 Future Work

- Collect and label field photographs from local farms, particularly for tomato and potato, and use them for training and evaluation.
- Fine-tune the full backbone on GPU hardware and evaluate larger architectures such as EfficientNet.
- Add an object-detection model such as YOLO [33] to localise individual pests and lesions.
- Learn severity and segmentation from annotated lesion masks instead of colour rules.
- Run the classifier on the phone for offline use, with the server providing reports and history when connectivity is available.
- Integrate IoT soil and weather sensors for continuous monitoring and automatic alerts, and aggregate diagnoses into regional disease maps.
- Provide multilingual and voice interfaces for farmers.

CHAPTER 9

Conclusion

This project designed, implemented and evaluated AgriVision AI, a crop health diagnosis system that turns a single photograph of a leaf into a diagnosis, a severity assessment and context-aware recommendations. The system combines image validation, two complementary leaf-segmentation strategies, a MobileNetV3-Large classifier trained by transfer learning on laboratory and field images, calibrated confidence with abstention, rule-based severity and nutrient screening, and a knowledge-based recommendation engine that considers crop, severity, growth stage, weather, soil readings, region and field history. It is delivered through a web application, a REST API and an Android application, with PDF reports and persistent history, and runs entirely on a laptop CPU.

On laboratory images, the classifier reaches 97.0% accuracy and, with abstention, a false diagnosis rate of 1.81%. On real field photographs, including field images in training raised accuracy from 19.9% to 53.8%, with 73.7% top-3 accuracy and 89.8% category accuracy. These results show both what the approach achieves and where its limits lie: laboratory accuracy alone is not a reliable indicator of field performance, and a practical tool must communicate its uncertainty and decline to answer when it is unsure. AgriVision AI is therefore positioned as an early-screening and decision-support tool that brings a first diagnosis to the farmer within seconds, while high-impact decisions remain subject to confirmation by an agricultural expert or laboratory testing.

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Appendix A Per-Class Results

Precision, recall and F1-score on the laboratory test set, and accuracy on the field test set, for each class. Field accuracy is not available for classes without PlantDoc test images.

Table 14: Per-class results on the laboratory and field test sets

Class	Precision	Recall	F1	*n* (lab)	Field acc.	*n* (field)
Apple - Apple Scab	0.967	0.983	0.975	60	0.40	10
Apple - Black Rot	1.000	1.000	1.000	60	—	—
Apple - Cedar Apple Rust	0.976	0.976	0.976	41	0.70	10
Apple - Healthy	0.967	0.983	0.975	60	0.56	9
No leaf detected	1.000	1.000	1.000	60	—	—
Blueberry - Healthy	1.000	1.000	1.000	60	0.45	11
Cherry - Powdery Mildew	1.000	1.000	1.000	60	—	—
Cherry - Healthy	1.000	1.000	1.000	60	0.30	10
Maize - Gray Leaf Spot	0.869	0.883	0.876	60	0.25	4
Maize - Common Rust	1.000	1.000	1.000	60	0.90	10
Maize - Northern Leaf Blight	0.881	0.867	0.874	60	0.58	12
Maize - Healthy	1.000	1.000	1.000	60	—	—
Grape - Black Rot	1.000	0.950	0.974	60	0.62	8
Grape - Esca (Black Measles)	0.952	1.000	0.976	60	—	—
Grape - Leaf Blight (Isariopsis Leaf Spot)	1.000	1.000	1.000	60	—	—
Grape - Healthy	1.000	1.000	1.000	60	1.00	12
Citrus - Huanglongbing (Citrus Greening)	1.000	1.000	1.000	60	—	—
Peach - Bacterial Spot	1.000	0.950	0.974	60	—	—
Peach - Healthy	0.964	1.000	0.982	54	0.67	9
Bell Pepper - Bacterial Spot	1.000	0.933	0.966	60	0.22	9
Bell Pepper - Healthy	0.923	1.000	0.960	60	0.88	8
Potato - Early Blight	0.968	1.000	0.984	60	0.50	8
Potato - Late Blight	1.000	0.967	0.983	60	0.25	8
Potato - Healthy	0.920	1.000	0.958	23	—	—
Raspberry - Healthy	1.000	1.000	1.000	56	1.00	7
Soybean - Healthy	0.984	1.000	0.992	60	0.50	8
Squash - Powdery Mildew	1.000	1.000	1.000	60	1.00	6

Class	Precision	Recall	F1	*n* (lab)	Field acc.	*n* (field)
Strawberry - Leaf Scorch	1.000	0.983	0.992	60	—	—
Strawberry - Healthy	1.000	1.000	1.000	60	1.00	8
Tomato - Bacterial Spot	0.950	0.950	0.950	60	0.11	9
Tomato - Early Blight	0.852	0.867	0.860	60	0.33	9
Tomato - Late Blight	0.982	0.917	0.948	60	0.40	10
Tomato - Leaf Mold	0.983	0.950	0.966	60	0.17	6
Tomato - Septoria Leaf Spot	0.981	0.883	0.930	60	0.45	11
Tomato - Spider Mite Damage	0.873	0.917	0.894	60	—	—
Tomato - Target Spot	0.875	0.933	0.903	60	—	—
Tomato - Tomato Yellow Leaf Curl Virus	0.983	0.983	0.983	60	0.83	6
Tomato - Tomato Mosaic Virus	0.982	1.000	0.991	56	0.10	10
Tomato - Healthy	0.983	0.967	0.975	60	0.38	8

Appendix B Knowledge Base Entry

The knowledge-base entry for tomato early blight, as used by the recommendation engine:

Table 15: Knowledge-base entry for tomato early blight

Field	Content
Crop / condition / category	Tomato / Early Blight / disease (fungal)
Cause	Fungal infection (Alternaria solani)
Symptoms	• Circular brown lesions with concentric rings • Yellowing around the affected regions • Lower, older leaves affected first
Cultural actions	• Remove severely affected leaves • Avoid prolonged leaf wetness • Improve plant spacing and airflow • Monitor nearby plants
Treatment	Apply a locally approved fungicide (e.g. chlorothalonil or mancozeb)
Prevention	• Mulch to stop soil splashing onto leaves • Stake plants and rotate crops for 2-3 years
Favourable conditions	24–29 °C, relative humidity ≥ 70%: warm, humid weather with long leaf wetness